

Joohyun Lee

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RESEARCH INTERESTS

Organic Electrochemical Devices (OECTs, OECDs), Conjugated Polymer Synthesis, Bioelectronic Applications

EDUCATION

Chung-Ang University (CAU)

- M.S. in Chemical Engineering
GPA: 4.18 / 4.5
- B.S. in Chemical Engineering and Materials Science
GPA: 3.75 / 4.5
Leave of absence for mandatory military service (May. 2019–Dec. 2020)

Seoul, Republic of Korea

Sep. 2024–Present

Advisor: Prof. Felix Sunjoo Kim

Mar. 2018–Aug. 2024

Advisor: Prof. Felix Sunjoo Kim

PUBLICATIONS

1. “Morphology and Crystallinity of Transparent Capacitive Ion Storage Layer for Electrochromic Displays.”
 - Seungwon Lee, Yuseok Song, **Joohyun Lee**, Jaeyong Ahn, Won-June Lee, Jiao Suo, Yoonhoo Lee, Liyan You, Jianguo Mei, Felix Sunjoo Kim, and Inho Song*.
 - *ACS Applied Materials & Interfaces*, accepted (2026).
2. “Enhancement-Mode PEDOT:PSS Electrochemical Transistors with Solid Electrolytes Based on [DBUH][OAc] as a Dual-Role Ionic Component.”
 - Donguk Kim[†], **Joohyun Lee**[†], Seungjin Song, Youngseok Kim, and Felix Sunjoo Kim*.
 - Under review
3. “Electrolyte-Driven Threshold Voltage Modulation in PEDOT:PSS OECTs via Anion Control of DBU-based Acid-Base Ionic Liquids.”
 - **Joohyun Lee**[†], Donguk Kim[†], Youngseok Kim, Seungjin Song, and Felix Sunjoo Kim*.
 - Manuscript in preparation; submission expected in 2026.
4. “Systematic Tuning of Energy Levels in EDOT-Based D-A-D Monomers via Acceptor Engineering and Oxidative Polymerization.”
 - **Joohyun Lee**, Seungjin Song, and Felix Sunjoo Kim*.
 - In preparation.

RESEARCH EXPERIENCE

Prof. Felix Sunjoo Kim Lab, Chung-Ang University

Graduate Researcher

Seoul, Republic of Korea

Sep. 2024–Present

- **Project 1:** Systematic Tuning of Energy Levels in EDOT-Based D-A-D Monomers via Acceptor Engineering and Oxidative Polymerization
 - Synthesized EDOT-based D-A-D monomers via Stille coupling (confirmed by ¹H NMR) and exploited the high reactivity of the α -site of EDOT for facile oxidative polymerization via aqueous-phase (PSS-templated) and surface methods, adding a base to retard the rapid polymerization during spin-coating
 - Determined the energy levels of synthesized monomers and polymers by correlating optical bandgap (UV-Vis-NIR) with electrochemical measurements (CV) and DFT calculations (ORCA)
- **Project 2:** Electrolyte-Driven Threshold Voltage Modulation in PEDOT:PSS OECTs via Anion Control of DBU-based Acid-Base Ionic Liquids
 - Controlled the ionicity of DBU-based ABILs by systematically varying the pKa of carboxylic acid components used in synthesis, characterized via ¹H NMR spectroscopy and DFT calculations
 - Characterized threshold voltage shift in PEDOT:PSS OECTs as a function of ABIL ionicity-dependent de-doping level via chemical (UV-Vis-NIR, Raman, XPS), electrical (transfer/output curves), and electrochemical (CV, EIS) characterization
 - Implemented zero- V_{GS} -load OECT inverters by exploiting the ABIL-tuned threshold voltage

Prof. Felix Sunjoo Kim Lab, Chung-Ang University

Undergraduate Researcher

Seoul, Republic of Korea

Mar. 2022–Aug. 2024

- **Project 1:** Enhancement-Mode PEDOT:PSS Electrochemical Transistors with Solid Electrolytes Based on [DBUH][OAc] as a Dual-Role Ionic Component
 - Demonstrated that [DBUH][OAc] exhibits distinct interaction with PEDOT:PSS compared to [EMIM][TFSI], which is critical for enhancement-mode OECT operation

- Revealed via DFT and XPS that [DBUH][OAc] preferentially interacts with the ion-conduction pathway in dry PEDOT:PSS, yielding C-V profiles with distinct accumulation and depletion regions analogous to organic semiconductors
- **Project 2: Long-Term Memory Organic Transistor Utilizing Electrolyte and Ferroelectric Polymer Bilayer Gate Dielectric**
 - Demonstrated synaptic memory with >24-hour retention and lower operating voltage in bilayer-gated organic transistors compared to ferroelectric polymer-only counterparts
 - Initially attributed the memory to ferroelectric polarization based on P-E hysteresis of the bilayer, but revealed through control experiments without the ferroelectric polymer that the long-term memory originated from the increased ionic resistance of the low-ion-concentration electrolyte (in collaboration with H. Yang)

FELLOWSHIPS, AWARDS & HONORS

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| • BK21 FOUR Fellowship , Ministry of Education, Korea (monthly stipend, ~\$800) | Sep. 2024–Aug. 2026 |
| • Research Assistantship , CAU (full tuition waiver, ~\$6,000/semester) | Fall 2025 |
| • Pilot Teaching Assistant Scholarship , CAU | Sep. 2024–Aug. 2025 |
| • Academic Excellence Scholarship , CAU (ranked 1st in department) | Spring 2024 |
| • Undergraduate Research Scholarship , CAU (competitive selection) | Sep. 2022–Feb. 2024 |
| • Leadership Scholarship , CAU (American Football Club president & captain) | Mar. 2021–Feb. 2023 |

SELECTED CONFERENCES

1. "Switching from Depletion- to Enhancement-mode in PEDOT:PSS-based Electrochemical Transistors."
Donguk Kim, **Joohyun Lee**, Jihoon Kim, and Felix Sunjoo Kim*.
ITC 2024, 18th International Thin-Film Transistor Conference, 2024.
2. "Threshold Voltage Modulation of PEDOT:PSS Based Electrolyte-Gated Transistors by Anion Control in Acid-Base Type Ionic Liquid."
Joohyun Lee, Donguk Kim, Youngseok Kim, and Felix Sunjoo Kim*.
FPI 16 Symposium, 16th International Symposium on Functional π -Electron Systems, 2025.
3. "Non-Volatile Switching Characteristics of Organic Electrochemical Transistors Enabled by Extremely Low Ion Concentration of Gel Electrolyte."
Heewon Yang, Jungmin Lee, **Joohyun Lee**, Minuk Kim, Hyejoon Park, Donguk Kim, Youngseok Kim*, and Felix Sunjoo Kim*.
FPI 16 Symposium, 16th International Symposium on Functional π -Electron Systems, 2025.
4. "Ionicity of Protic Ionic Liquid Influences Channel Conductance of Conducting Polymer in Electrochemical Transistors."
Joohyun Lee, Donguk Kim, and Felix Sunjoo Kim*.
2026 KICChE Spring Meeting, 2026.
5. "Systematic Tuning of Energy Levels in EDOT-Based D-A-D Monomers via Acceptor Engineering."
Joohyun Lee, Seungjin Song, and Felix Sunjoo Kim*.
KJF-ICOMEF 2026, 2026 KJF-International Conference on Organic Materials for Electronics and Photonics, 2026.

SKILLS

- **Device & Circuit Characterization:** OECT fabrication and figure-of-merit extraction (g_m , V_T , on/off, μC^* , transient response); OECT inverter characterization (gain, noise margins) [HP4156A, Ki2634B]
- **Electrochemical Analysis:** CV, EIS, EIS-based C–V (C^*), Spectroelectrochemistry [SP-150e]
- **Synthesis:** Pd-catalyzed Stille coupling, oxidative polymerization (solution & surface), protic ionic liquid synthesis, column chromatography
- **Structural & Spectroscopic:** XPS, Raman, UV-Vis-NIR, FT-IR, NMR
- **Morphological & Thermal:** AFM, TGA, DSC
- **Computational:** DFT/TD-DFT (ORCA, Spartan), Python, MATLAB
- **Languages:** Korean (Native), English (Proficient)

ACTIVITIES

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| • President & Captain , CAU American Football Club | Mar. 2018–Feb. 2023 |
| Team member from 2018; president and captain (2021–2023), leading club operations and on-field performance. Seoul Collegiate American Football Tournament: Champion (2018), Runner-up (2019), 3rd Place (2021) | |

REFERENCES

Dr. Felix Sunjoo Kim, Professor, Chung-Ang University
Dr. Inho Song, Assistant Professor, Chung-Ang University

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